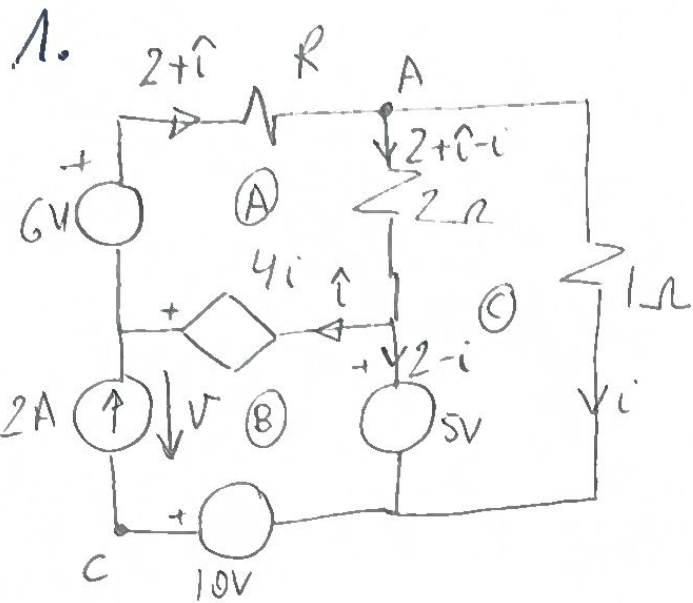




$$P_{2A} = 46W = 2 \cdot V \Rightarrow V = 23V$$

$$(B) \rightarrow 4i + 5 - 10 - V = 0 \Rightarrow i = 7A$$

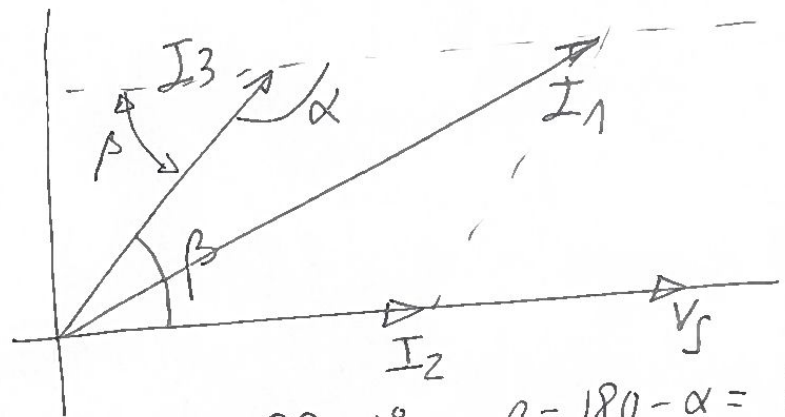
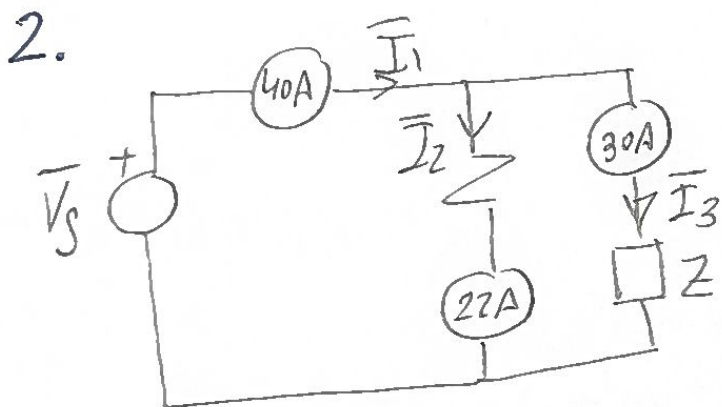
$$(C) \rightarrow 2(2 + \hat{i} - i) + 5 = i \Rightarrow \hat{i} = 6A$$

$$(A) \rightarrow 6 + 4i = R(2 + \hat{i}) + 2(2 + \hat{i} - i) \Rightarrow R = 4\Omega$$

$$V_{AB} = 2(2 + \hat{i} - i) + 5 - 10 \Rightarrow V_{AB} = -3V$$

$$P_{10V} = 10 \cdot 2 = 20W \text{ (gen)} \quad P_{6V} = 6(2 + \hat{i}) = 48W \text{ (gen)}$$

$$P_{5V} = 5(2 - i) = -25W \rightarrow 25W \text{ (gen)} \quad P_{4i} = 4i \cdot \hat{i} = 168W \text{ (gen)}$$



$$|I_1|^2 = |I_2|^2 + |I_3|^2 - 2|I_2||I_3|\cos\alpha \Rightarrow \alpha = 99,41^\circ$$

$$\beta = 180 - \alpha = 80,58^\circ$$

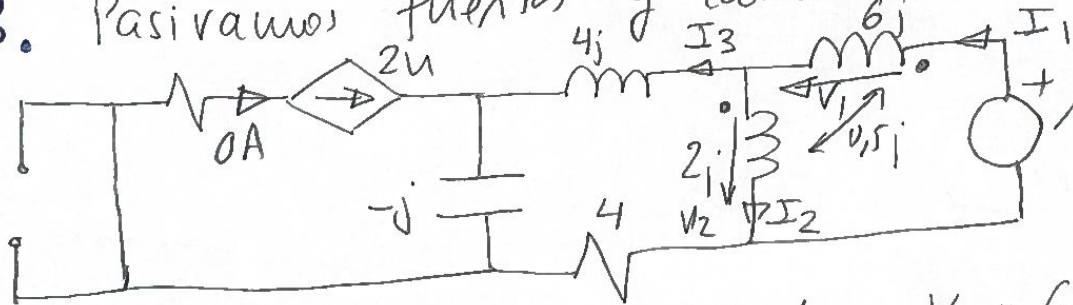
$$|V_s| = 22 \cdot 5 = 110V \quad |Z| = |V_s| / |I_3| = 3,667\Omega$$

$$Z = 3,667 \angle -80,58^\circ = 0,6 - 3,617j$$

$$P_g = 5 \cdot 22^2 + 0,6 \cdot 30^2 \Rightarrow P_g = 2960W$$

$$Y = \frac{1}{Z} = 0,0446 + 0,269j \quad Y_L = -0,269j = \frac{1}{\omega L} \Rightarrow L = 11,83mH$$

3. Pasivamos fuentes y calculamos Z_{th} .



$$u = 0V \Rightarrow 2u = 0A$$

$$Z_C = \frac{1}{\omega C} = -j$$

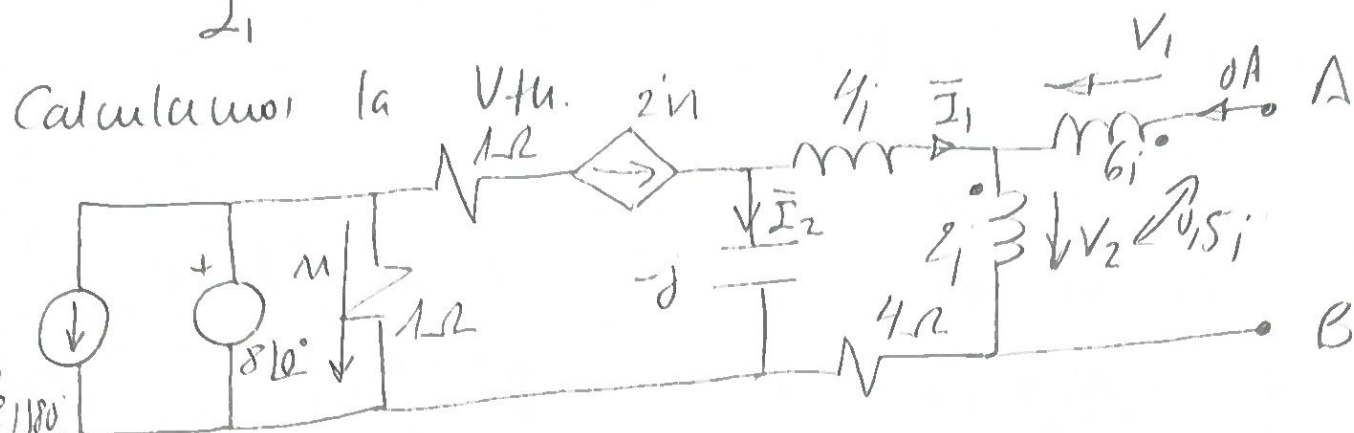
Emisiones bobinas acopladas: $V_1 = 6jI_1 + 0,5jI_2 \quad V_2 = 2jI_2 + 0,5jI_1$

Emisiones de malla:

$$1 = V_1 + V_2 \Rightarrow 6,5jI_1 + 2,5jI_2 = 1$$

$$1 = V_1 + (4 + 3j)(I_1 - I_2) \Rightarrow 1 = (4 + 9j)I_1 - (4 + 2,5j)I_2 \Rightarrow \begin{cases} I_1 = 0,0089 - 0,12j \\ I_2 = 0,02 - 0,08j \end{cases}$$

$$Z_{th} = \frac{1}{I_1} \Rightarrow Z_{th} = 0,6098 + 8,2378j \quad |Z_{max} = 0,6098 - 8,2378j|$$



$$M = 8 \angle 0^\circ \Rightarrow Z_n = 16 \angle 0^\circ$$

$$I_1 = 16 \frac{-j}{4j + 2j + 4j} = -1,9512 - 1,56j$$

$$I_2 = 16 \angle 0^\circ - I_1 = 17,95 + 1,56j$$

$$V_{th} = V_1 + V_2 = 0,5j I_1 + 2j I_2 \Rightarrow V_{th} = -3,902 + 4,87j \Rightarrow V_{th} = 6,24 \angle 128,7^\circ$$

$$P_{max} = \frac{(6,24)^2}{4 \cdot 0,6098} \Rightarrow |P_{max} = 15,96W|$$

4. $\int P_3 = W_1 + W_2 = 3000W$ $\cos \phi_3 = 0,822 \text{ (ind)}$
 $C_3 \left\{ \begin{aligned} Q_3 &= \sqrt{3} (W_1 - W_2) = 20784,60 \text{ VAR (ind)} \end{aligned} \right.$

$$P_T = 3W_3 = 56445W$$

$$Q_T = P_T \tan(\arccos 0,85) = 34981,4 \text{ var} \quad (*)$$

$$P_1 = 3 \frac{(380/\sqrt{3})^2}{10} = 14440W \quad Q_1 = 0 \text{ var}$$

$$P_2 = P_T - P_1 - P_3 = 12005W$$

$$Q_2 = Q_T - Q_1 - Q_3 = 14196,8 \text{ var}$$

$$\left. \begin{aligned} P_2 &= 12005W \\ Q_2 &= 14196,8 \text{ var} \end{aligned} \right\} \cos \phi_2 = 0,645 \text{ (ind)}$$

$$C^A = \frac{P_T (\tan(\arccos 0,85) - \tan(\arccos 0,95))}{2\pi f (380)^2} = 0,362 \text{ mF}$$

$$C^A = \frac{C^A}{3} \Rightarrow |C^A = 0,12 \text{ mF}|$$

Al conectar el condensador:

$$P_T' = P_T = 56445W = S_T \cdot 0,95 = \sqrt{3} \cdot 380 \cdot |I_2| \cdot 0,95 \Rightarrow |A_2 = 90,27A|$$

$$(*) \rightarrow S_T = \sqrt{P_T^2 + Q_T^2} = 66405,63 \text{ VA} = \sqrt{3} \cdot 380 \cdot A_1 \Rightarrow |A_1 = 100,89A|$$